

```
karthik@Karthikeya:~$ nano command_runner.c
karthik@Karthikeya:~$ gcc command_runner.c -o command_runner
karthik@Karthikeya:~$ ./command_runner
Enter Linux command: ls
```

```
[Parent Process] PID: 1935
[Child Process] PID: 1938 (Parent PID: 1935)
[Child Process] Executing: 'ls'...
```

File1.txt	MyFifo	command_executor	command_runner.c	process	process_states	task1.c	task3	task4.c	week3
File2.txt	OSSP	command_executor.c	fork_example.c	process.c	process_states.c	task2	task3.c	task5	
File3.txt	Week4	command_runner	name.txt	process1.c	task1	task2.c	task4	task5.c	

```
[Parent Process] Child process (PID: 1938) completed.
```

```
Marthik@Marthiksys:~$ cat /etc/os-release
Linux Marthiksys 5.15.13.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2024 x86_64 x86_64 GNU/Linux
Marthik@Marthiksys:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 24
On-line CPU(s) list:   0-23
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i7-13700HX
CPU family:             6
Model:                 191
Thread(s) per core:     2
Core(s) per socket:     12
Socket(s):              1
Stepping:               2
BogoMIPS:               4687.99
Flags:                  fpu vme de pss tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr ss
e sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable
nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2api
c movbe popcnt tsc_deadline_timer aes xsave avx f16c rdtrd hypervisor lahf_lm aba 3dnowprf
fetch_sdb idrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bti
1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv
1 xsave avx_vnni vnni umip waitpkg gdtl vmas vpcmlmulqdq rdpid movdiri movdir64b fsrm wrnq_c
ear serialize ibt flush_l1d arch_capabilities

Virtualization features:
Virtualization:         VT-x
Hypervisor vendor:      Microsoft
Virtualization type:     full
Caches (sum of all):
  L1d:                   576 KiB (12 instances)
  L1i:                   384 KiB (12 instances)
  L2:                    15 MiB (12 instances)
  L3:                    30 MiB (1 instance)
NUMA:
NUMA node(s):           1
NUMA node0 CPU(s):      0-23
Vulnerabilities:
  Gather data sampling:   Not affected
  Ghostwrite:            Not affected
  Indirect target selection: Not affected
  Itlb multihit:         Not affected
  L1tf:                  Not affected
  Mds:                   Not affected
  Meltdown:              Not affected
  Mmio stale data:       Not affected
  Old microcode:         Not affected
  Rng file data sampling: Mitigation; Clear Register File
  Retbleed:              Mitigation; Enhanced IBRS
  Spec rstack overflow:   Not affected
  Spec store bypass:     Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:            Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:            Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSB-IBRS SW sequence; BMI BMI_D
IS_5
  Srbds:                 Not affected
  Tsx:                   Not affected
  Tsx async abort:       Not affected
  Untrn:                 Not affected
```

```
karthik@karthikeya:~$ lsblk
NAME
MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 1 disk
sdb 8:16 0 159.4M 1 disk
sdc 8:32 0 2G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro
```

```
karthik@karthikeya:~$ ps
PID TTY TIME CMD
308 pts/0 00:00:00 bash
1944 pts/0 00:00:00 ps
```

```
karthik@karthikeya:~$ top
top - 15:42:50 up 1:19, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7791.7 total, 7081.3 free, 609.8 used, 258.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7181.8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1945	karthik	20	0	9284	5728	3492	R	33.3	0.1	0:00.61	top
1	root	20	0	21724	12992	9624	S	0.0	0.2	0:00.73	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
6	root	20	0	3196	2128	2008	S	0.0	0.0	0:00.00	init
52	root	19	-1	66836	16004	14796	S	0.0	0.2	0:00.74	systemd-journal
99	root	20	0	25016	6404	5168	S	0.0	0.1	0:00.92	systemd-udevd
108	systemd+	20	0	21344	13220	11096	S	0.0	0.2	0:00.13	systemd-resolve
109	systemd+	20	0	91036	7956	6988	S	0.0	0.1	0:00.26	systemd-timesyn
163	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.01	cron
164	message+	20	0	9600	5420	4708	S	0.0	0.1	0:00.25	dbus-daemon
176	root	20	0	17984	8884	7848	S	0.0	0.1	0:00.13	systemd-logind
179	root	20	0	1756104	12024	9268	S	0.0	0.2	0:00.31	wsl-pro-service
188	root	20	0	3124	1968	1832	S	0.0	0.0	0:00.00	agetty
190	syslog	20	0	222516	5952	4556	S	0.0	0.1	0:00.20	rsyslogd
208	root	20	0	107032	23168	13712	S	0.0	0.3	0:00.06	unattended-upgr
306	root	20	0	3188	1116	980	S	0.0	0.0	0:00.00	SessionLeader
307	root	20	0	3204	1252	1108	S	0.0	0.0	0:00.53	Relay(308)
308	karthik	20	0	6596	5920	3672	S	0.0	0.1	0:00.13	bash
309	root	20	0	6704	4672	3888	S	0.0	0.1	0:00.00	login
353	karthik	20	0	20324	11560	9440	S	0.0	0.1	0:00.12	systemd
354	karthik	20	0	21168	3576	1848	S	0.0	0.0	0:00.00	(sd-pam)
380	karthik	20	0	6080	5332	3660	S	0.0	0.1	0:00.01	bash
733	polkitd	20	0	308172	8128	7228	S	0.0	0.1	0:00.20	polkitd
1946	root	20	0	25020	3548	2280	S	0.0	0.0	0:00.00	(udev-worker)
1947	root	20	0	25020	3548	2280	S	0.0	0.0	0:00.00	(udev-worker)